

Statistics MCQ Question Bank

First Paper

Abdullah Al Mahmud

Last updated: September 29, 2026

www.statmania.info

Contents

1 Basic Concept of Statistics

1. **Who is known as the Father of modern statistics?**
(a) P.C. Mahalanobis (b) Kazi Motaher Hos-sain (c) Karl Pearson (d) R.A. Fisher
2. **Which is not a function of statistics?**
(a) Data collection (b) Data organization (c) Analysis (d) Database creation
3. **Which one is an example of an infinite population?**
(a) Students of Dhaka University (b) Cadets of SCC
(c) Minor planets in the solar system (d) Red blood cells in a person's body
4. **Which of the following is an example of an infinite population?**
(a) Employees of a multinational company (b) Trees in a national park
(c) Stars in the Milky Way (d) Passengers on a flight
5. **Which one represents an infinite population?**
(a) Books in a library (b) Fish in the Pacific Ocean
(c) Members of a sports club (d) Mobile phones in a city
6. **A researcher collected data on age and income of the people in a city. The variables are –**
i. bi-variate
ii. quantitative
iii. qualitative
Which one is correct?
(a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii
7. **Which of the following is correct?**
(a) $\sum_{i=1}^{20} cx_i = nc \sum_{i=1}^{20} x_i$ (b) $\sum_{i=1}^{20} cx_i = nc \sum_{i=1}^{20} x_i$ (c) $\sum_{i=1}^{20} cx_i = c \sum_{i=1}^{20} x_i$ (d) $\sum_{i=1}^{20} cx_i = c^2 \sum_{i=1}^{20} x_i$
8. **Which cannot be performed using Univariate data?**
(a) Central tendency (b) Dispersion (c) Skewness (d) Regression
9. **Which of the following cannot be analyzed using univariate data?**
(a) Mean (b) Variance (c) Correlation (d) Range
10. **Which statistical method requires bivariate or multivariate data?**
(a) Standard deviation (b) Histogram (c) Regression analysis (d) Median
11. **Which of the following is an example of an infinite population?**
(a) Patients in a hospital (b) Water molecules in the ocean
(c) Cars on a highway (d) Students in a university
12. **Which of the following is an example of a finite population?**
(a) Books in a school library (b) Stars in the universe
(c) Grains of sand on a beach (d) Atoms in the atmosphere

13. Which one represents an infinite population?
- (a) Trees in a forest (b) Grains of sand on a beach
(c) Books in a bookstore (d) Houses in a neighborhood
14. Cities ranked according to habitability level show – measurement scale
- (a) Nominal (b) Ratio (c) Interval (d) Ordinal
15. Classifying students based on their grades (A, B, C, etc.) represents which measurement scale?
- (a) Nominal (b) Ordinal (c) Interval (d) Ratio
16. Temperature measured in Celsius or Fahrenheit follows which type of measurement scale?
- (a) Nominal (b) Ordinal (c) Interval (d) Ratio
17. A survey categorizing people by their favorite color is an example of which measurement scale?
- (a) Nominal (b) Ordinal (c) Interval (d) Ratio
18. Which is not an example of shift of scale?
- (a) $y_i = \frac{x_i}{a}$ (b) $y_i = cx_i$ (c) $y_i = x_i - 2$ (d) $y_i = \frac{cx_i}{d}$
19. If $\sum_{i=1}^{20} x_i^2 = 20$ and $\sum_{i=1}^{20} x_i = 30$, what is the value of $\sum_{i=1}^{20} x_i^2 + \sum_{i=1}^{20} x_i + 100$?
- (a) 130 (b) 200 (c) 150 (d) 2130
20. If $\sum_{i=1}^{15} y_i^2 = 50$ and $\sum_{i=1}^{15} y_i = 25$, what is the value of $\sum_{i=1}^{15} y_i^2 - \sum_{i=1}^{15} y_i + 75$?
- (a) 100 (b) 50 (c) 125 (d) 45
21. Given $\sum_{i=1}^{10} a_i^2 = 40$ and $\sum_{i=1}^{10} a_i = 20$, find the value of $2 \sum_{i=1}^{10} a_i^2 - 3 \sum_{i=1}^{10} a_i + 60$.
- (a) 70 (b) 100 (c) 80 (d) 50
22. If $\sum_{i=1}^{25} z_i^2 = 75$ and $\sum_{i=1}^{25} z_i = 50$, compute $\sum_{i=1}^{25} z_i^2 + 2 \sum_{i=1}^{25} z_i - 125$.
- (a) 50 (b) 75 (c) 100 (d) 25
23. A subset of a population is called–
- (a) Constant (b) Variable (c) Sample (d) Scale
24. What is $\sum_{i=1}^n bx_i$ equal to?
- (a) $b \sum_{i=1}^n nx_i$ (b) $b \sum_{i=1}^n x_i$ (c) $\sum_{i=1}^n nx_i$ (d) $bn \sum_{i=1}^n x_i$
25. How many measurement scales are there?
- (a) 2 (b) 3 (c) 4 (d) 5

26. **Which of the following is a continuous variable?**
 (a) Number of goals (b) Natural number
 (c) Summation of Fibonacci series (d) Success rate
27. **In which scale of measurement, zero is regarded as true zero?**
 (a) Nominal scale (b) Interval scale (c) Ratio scale (d) Ordinal scale
28. **Which measurement scale does height belong to?**
 (a) Nominal (b) Ordinal (c) Interval (d) Ratio
29. **Which is a discrete variable?**
 (a) Weight (b) Amount of rainfall (c) Distance (d) Grade in a subject
30. **Which is a discrete variable?**
 (a) Height of a building (b) Number of cars in a parking lot
 (c) Amount of milk in a container (d) Time taken to complete a task
31. **Which is a discrete variable?**
 (a) Speed of a car (b) Number of students in a class
 (c) Volume of water in a tank (d) Temperature of a room
32. **Which is a discrete variable?**
 (a) Blood pressure (b) Number of books on a shelf
 (c) Length of a river (d) Amount of sugar in a cup
33. **Which is a discrete variable?**
 (a) Shoes sizes available in a store (b) Distance between two cities
 (c) Volume of a gas (d) Weight of a parcel
34. **Which is a discrete variable?**
 (a) Grades on a multiple-choice test (A, B, C, D) (b) Temperature during the day
 (c) Height of a person (d) Time spent on an activity
35. **Which is a discrete variable?**
 (a) Outcomes of rolling a die (b) Speed of a train
 (c) Rainfall in a region (d) Age of a tree
36. **Which is a discrete variable?**
 (a) Counts of people in a room (b) Temperature recorded every hour
 (c) Weight of an animal (d) Height of a plant
37. **Which is a discrete variable?**
 (a) Number of languages spoken by a person (b) Time taken to complete a race
 (c) Length of a road (d) Volume of water in a tank
38. **Which is a discrete variable?**
 (a) Length of a rope (b) Weight of books in a library
 (c) Distance (d) No. of particles in atoms

39. If $x_1 = 2$, $x_2 = -3$, $x_3 = 7$, and $x_4 = 12$, $\sum_{i=1}^4 x_i^2 = ?$
- (a) 26 (b) 106 (c) 206 (d) 216
40. If $x_1 = 5$, $x_2 = -4$, $x_3 = 9$, and $x_4 = 0$, what is $\sum_{i=1}^4 x_i^2$?
- (a) 82 (b) 97 (c) 107 (d) 122
41. If $x_1 = 3$, $x_2 = 2$, $x_3 = -6$, and $x_4 = 4$, what is $\sum_{i=1}^4 x_i^2$?
- (a) 45 (b) 65 (c) 85 (d) 89
42. If $x_1 = 4$, $x_2 = 1$, $x_3 = -2$, and $x_4 = 3$, find $\sum_{i=1}^4 (x_i^2 + 3)$?
- (a) 40 (b) 50 (c) 42 (d) 56
43. If $y_1 = 5$, $y_2 = 2$, $y_3 = -1$, and $y_4 = 4$, $\sum_{i=1}^4 (y_i^2 + 2) = ?$
- (a) 50 (b) 40 (c) 54 (d) 60
44. Given $z_1 = 3$, $z_2 = 0$, $z_3 = -3$, and $z_4 = 2$, what is the value of $\sum_{i=1}^4 (z_i^2 + 5)$?
- (a) 30 (b) 40 (c) 35 (d) 45
45. If $x_1 = 4$, $x_2 = -2$, $x_3 = 1$, and $x_4 = 5$, calculate $\sum_{i=1}^4 (2x_i^2 - x_i)$?
- (a) 38 (b) 42 (c) 46 (d) 84
46. If $x_1 = 3$, $x_2 = 1$, $x_3 = 0$, and $x_4 = 2$, find $\sum_{i=1}^4 x_i^2 - \sum_{i=1}^4 x_i$?
- (a) 7 (b) 9 (c) 8 (d) 13
47. If $x_1 = 5$, $x_2 = 4$, $x_3 = -3$, and $x_4 = 2$, find $\sum_{i=1}^4 (x_i^2 + x_i)$?
- (a) 58 (b) 62 (c) 66 (d) 72
48. If $x_1 = 2$, $x_2 = 3$, $x_3 = -1$, and $x_4 = 0$, calculate $\sum_{i=1}^4 (x_i^2 - 2)$?
- (a) 0 (b) 6 (c) 8 (d) 10
49. If $x_1 = 2$, $x_2 = 3$, $x_3 = 4$, $x_4 = 6$, and $x_5 = 5$, $\sum_{i=1}^4 x_i^2 = ?$
- (a) 80 (b) 87 (c) 90 (d) 105

50. If $f_i = 3, 5, 7$ and $x_i = 2, 4, 7$; what is the value of $\sum_{i=1}^3 f_i x_i^2$?
- (a) 450 (b) 350 (c) 345 (d) 435
51. If $f_i = 2, 4, 6$ and $x_i = 3, 5, 7$, what is the value of $\sum_{i=1}^3 f_i x_i^3$?
- (a) 950 (b) 1125 (c) 2612 (d) 1330
52. Given $f_i = 1, 3, 5$ and $x_i = 2, 4, 6$, find the value of $\sum_{i=1}^3 f_i x_i^4$.
- (a) 1356 (b) 1536 (c) 1650 (d) 7264
53. If $f_i = 3, 5, 7$ and $x_i = 2, 4, 6$, compute $\sum_{i=1}^3 f_i x_i^2$.
- (a) 260 (b) 280 (c) 344 (d) 320
54. Find the value of $\sum_{i=1}^{12} f_i (x_i - 7)^2$ where $\sum_{i=1}^{12} f_i x_i^2 = 400$, $\sum_{i=1}^{12} f_i x_i = 40$, $\sum_{i=1}^{12} f_i = 10$
- (a) 320 (b) 330 (c) 250 (d) 430
55. If $x_1 = 3$, $x_2 = -1$, $x_3 = 2$, and $x_4 = 0$, find $\sum_{i=1}^4 (x_i^3 + 2x_i)$?
- (a) 12 (b) 18 (c) 24 (d) 28
56. If $x_1 = 4$, $x_2 = 1$, $x_3 = -2$, and $x_4 = 3$, calculate $\sum_{i=1}^4 (x_i^2 + 4x_i - 1)$?
- (a) 16 (b) 24 (c) 34 (d) 50
57. If $x_1 = 1$, $x_2 = 2$, $x_3 = -3$, and $x_4 = 4$, find $\sum_{i=1}^4 (3x_i^3 - x_i^2)$?
- (a) 108 (b) 114 (c) -8 (d) 201
58. If $x_1 = 5$, $x_2 = 0$, $x_3 = -1$, and $x_4 = 2$, determine $\sum_{i=1}^4 (x_i^3 + x_i^2 + 3)$?
- (a) 173 (b) 174 (c) 164 (d) 172
59. Capital and profit belong to a variable which is—
- Bivariate
 - Quantitative
 - Qualitative
- Which one is correct?
- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii
60. Which one falls in the category of interval scale?
- (a) Temperature (b) Speed (c) Distance (d) Film rating

61. Which one falls in the category of nominal scale?
 (a) Height (b) Temperature (c) Gender (d) Age
62. Which of the following is an example of an ordinal scale?
 (a) Temperature (b) IQ Score (c) Educational Level (d) Weight
63. Which of the following is not example of a ratio scale?
 (a) Temperature (b) Time (c) Blood Pressure (d) Speed
64. In which scale of measurement, zero is regarded as true zero?
 (a) Nominal scale (b) Interval scale (c) Ratio scale (d) Ordinal scale
65. Which is a discrete variable?
 (a) Weight (b) Amount of rainfall (c) Distance (d) Grade in a subject
66. Which one is product of square?
 (a) $\prod x_i^2$ (b) $(\prod x_i)^2$ (c) $\sum x_i^2 \times \sum x$ (d) $\sum x_i^2$
67. For which variable, determining number of terms is not possible?
 (a) Discrete variable (b) Continuous variable (c) Quantitative variable (d) Qualitative variable
- Answer the next three question based on the following information.**

A farmer collects growth (in cm) of 10 plants in a month and finds that
 $\sum x_i = 7$ and $\sum x_i^2 = 15$

68. Which is considered statistics?
 (a) Jaman obtained 75 in statistics (b) Shafiq lives at Road no. 5
 (c) Mean monthly income in a city is 60,000 taka (d) Width of a book is 10 cm
69. What is the value of $\sum(x_i + 4)$ if $x = \{2, 3\}$?
 (a) 23 (b) 47 (c) 22 (d) 13
70. If $x_1 = 2, x_2 = 3, x_3 = 5, x_4 = 7$ and $y_1 = 3, y_2 = 4, y_3 = 5, y_4 = 8$; $\sum_{i=2}^4 x_i y_i = ?$
 (a) 14 (b) 201 (c) 93 (d) 117
71. From the following table, $\sum_{i=1}^4 x_i y_i = ?$

X	1	5	3	2
Y	20	12	3	14

- (a) 14 (b) 201 (c) 99 (d) 109
72. What is the value of $\sum(x_i - 4)^2$?
 (a) 23 (b) 135 (c) 484 (d) 119
73. If the square of summation is subtracted the sum of square, the value is -
 (a) -8 (b) 34 (c) 8 (d) -34

74. Which one is not an example of ratio scale?

- (a) Room no. (b) Income (c) Number of accidents (d) Weight

75. Which one is discrete?

- (a) Weight (b) Amount of rainfall
(c) Temperature (d) No. of member in a family

76. Which type of scale of measurement are religion and blood group?

- (a) Interval (b) Ratio (c) Nominal (d) Ordinal

Answer the next two questions based on the following information

$$X = 20, 25, 30, 40$$

77. Find $\sum(X_i + 10)$

- (a) 150 (b) 155 (c) 125 (d) 250

78. $\sum(X_i - 30)^2$

- (a) 225 (b) 230 (c) 420 (d) 235

Answer the next two questions based on the following information

$$X = 3, 5, 7, 10$$

79. Find $\sum(X_i + 3)$

- (a) 28 (b) 32 (c) 37 (d) 40

80. $\sum(X_i - 5)^2$

- (a) 16 (b) 33 (c) 12 (d) 8

Answer the next two questions based on the following information

$$X = 6, 8, 10, 12$$

81. Find $\sum(X_i - 4)$

- (a) 20 (b) 30 (c) 32 (d) 22

82. $\sum(X_i + 2)^2$

- (a) 196 (b) 504 (c) 210 (d) 220

Answer the next two questions based on the following information

$$X = 4, 9, 13, 15$$

83. Find $\sum(2X_i)$

- (a) 68 (b) 70 (c) 82 (d) 74

84. $\sum(X_i - 10)^2$

- (a) 71 (b) 80 (c) 85 (d) 92

Answer the next three questions based on the following information.

The values of x_i and f_i are given below:

x_i	1	2	3	4
f_i	2	3	4	1

85. Find $\sum_{i=1}^4 f_i x_i$.

- (a) 20 (b) 21 (c) 22 (d) 24

86. Compute $\sum_{i=1}^4 f_i x_i^2$.

- (a) 30 (b) 35 (c) 66 (d) 64

87. Determine $\sum_{i=1}^4 f_i^2 x_i$.

- (a) 74 (b) 49 (c) 78 (d) 65

Answer the next three questions based on the following information.

The values of x_i and f_i are given below:

x_i	2	4	6	8
f_i	2	2	5	4

88. Find $\sum_{i=1}^4 f_i x_i$.

- (a) 50 (b) 74 (c) 56 (d) 60

89. Compute $\sum_{i=1}^4 f_i x_i^2$.

- (a) 256 (b) 274 (c) 476 (d) 300

90. Determine $\sum_{i=1}^4 f_i (x_i - 5)^2$.

- (a) 61 (b) 48 (c) 52 (d) 58

1.1 Situation Set

Answer the next TWO questions based on the following information

The table below shows monthly sales (in thousands of dollars) for two products in four regions.

Product	North	South	East	West
A	12	15	10	8
B	18	22	14	16

91. What is the total sales of Product A across all regions?

- (a) 40 (b) 42 (c) 45 (d) 48

92. What is the total sales of all products across all regions?

- (a) 95 (b) 100 (c) 105 (d) 115

2 Collection, Organization, and Presentation of Data

93. How many sources of data are there?
(a) 5 (b) 4 (c) 3 (d) 2
94. What is the raw material of research?
(a) Data (b) Theory (c) Graph (d) Mean
95. Data obtained through direct observation is called—
(a) Primary data (b) Secondary data (c) Original Data (d) Informal data
96. Which formula is used to find angles for Pie Chart?
(a) $\theta_i = \frac{f_i}{N} \times 100$ (b) $\theta_i = \frac{f_i}{100} \times 360$ (c) $\theta_i = \frac{f_i}{N} \times 360$ (d) $\theta_i = \frac{f_i}{N-1} \times 360$
97. Who invented Stem and Leaf plot?
(a) Karl Pearson (b) R.A. Fisher (c) David Cox (d) John Tukey
98. If all the rats in Sylhet is a population, all the rats in Sylhet Airport is —
(a) Data (b) Sample (c) Statistics (d) Frequency
99. Which rule is suggested by H.G. Sturges for determining number of class (k)?
(a) $K = 1 + 3.322\log N$ (b) $K = 1 + 3.222\log N$ (c) $K = 1 - 3.222\log N$ (d) $K = 1 + 2.332\log N$
100. To show runs per over in a cricket match, which diagram can be used?
(a) Histogram (b) Bar Diagram (c) Ogive (d) Frequency polygon
101. Which of the following is most suitable for comparing the sales of five different products over a single year?
(a) Histogram (b) Pie Chart (c) Simple Bar Diagram (d) Frequency Curve
102. To display the trend of population growth over 50 years, the best choice is:
(a) Line Diagram (b) Histogram (c) Pie Chart (d) Scatter Plot
103. Which diagram is used to find the median graphically?
(a) Histogram (b) Bar Diagram (c) Ogive (d) Frequency Polygon
104. A pie chart is most appropriate for showing:
(a) Frequency distribution (b) Component parts of a total
(c) Trend over time (d) Correlation between two variables
105. To show the relationship between height and weight of 50 students, the recommended diagram is:
(a) Histogram (b) Scatter Plot (c) Bar Diagram (d) Pie Chart
106. A frequency polygon is drawn by joining the midpoints of the tops of rectangles in a:
(a) Pie Chart (b) Histogram (c) Ogive (d) Bar Diagram
107. To compare the rainfall of two cities over 12 months, an appropriate diagram is:
(a) Multiple Bar Diagram (b) Simple Bar Diagram (c) Pie Chart (d) Scatter Plot
108. Which graph is used to locate quartiles graphically?
(a) Histogram (b) Frequency Polygon (c) Ogive (d) Scatter Plot

2.1 Situation Set

Answer the next **THREE** questions based on the following information

Radius of 80 trees are recorded and this frequency distribution is constructed.

Radius (cm)	0-10	10-20	20-30	30-40
No. of Trees	20	15	21	24

109. How many trees have radius between 10 and 30?

- (a) 30 (b) 15 (c) 36 (d) 21

110. How many trees have radius at least 20?

- (a) 44 (b) 45 (c) 24 (d) 21

111. What percent of trees have radius between 20 and 40?

- (a) 44% (b) 56% (c) 46% (d) 53%

Answer the next two questions based on the following plot

Data: 18, 21, 22, 23, 24, 26, 31, 33, 33, 35, 37, 42

Stem	Leaf
1	8
2	1 2 3 4 6
3	1 3 3 5 7
4	2

Key: 2 — 1 means **21**

112. How many data values are greater than 30 in the stem-and-leaf plot?

- (a) 3 (b) 4 (c) 5 (d) 6

113. What is the median of the data shown in the stem-and-leaf plot?

- (a) 26 (b) 31 (c) 30 (d) 29

Answer the next two questions based on the following plot

Stem	Leaf
1	3 7 8
2	0 2 5 9
3	1 1 4 6
4	0 3

Key: 2 — 5 = 25

114. What is the mode of the data set?

- (a) 31 (b) 22 (c) 13 (d) None (no mode)

115. What is the median of the data shown in the stem-and-leaf plot?

- (a) 26 (b) 31 (c) 30 (d) 29

Answer the next **THREE** questions based on the following information.

The heights of 100 plants were measured, and this frequency distribution was constructed.

Height (cm)	0-20	20-40	40-60	60-80
No. of Plants	25	30	20	25

116. How many plants have height between 20 and 60?

- (a) 50 (b) 30 (c) 20 (d) 25

117. How many plants have height at least 40?

- (a) 50 (b) 45 (c) 40 (d) 25

118. What percent of plants have height between 20 and 80?

- (a) 80% (b) 75% (c) 60% (d) 50%

Answer the next THREE questions based on the following information.

The weights of 120 fruits were recorded and this frequency distribution was constructed.

Weight (grams)	0-50	50-100	100-150	150-200
No. of Fruits	30	35	25	30

119. How many fruits weigh at least 100 grams?

- (a) 55 (b) 50 (c) 60 (d) 65

120. How many fruits weigh less than 100 grams?

- (a) 68 (b) 70 (c) 65 (d) 50

121. What percent of fruits weigh between 50 and 150 grams?

- (a) 50% (b) 55% (c) 60% (d) 75%

Answer the next two questions based on the following information

Class Interval	0-10	10-20	20-30	30-40
Frequency	6	3	7	4

122. What is relative frequency of the class with the highest frequency?

- (a) 0.25 (b) 0.45 (c) 0.40 (d) 0.35

123. Which curve is suitable for

- (a) Histogram (b) Bar Diagram (c) Pie Chart (d) Ogive

124. Example of primary data —

- i. A student collected data for research
ii. A professor had a student collect data for them
iii. A researcher collected data from a newspaper.

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

125. Which of the following is an example of secondary data?

- i. Data obtained from a published journal
ii. Data collected by a government agency and used by a researcher
iii. Data gathered directly through interviews

Which one is correct?

- (a) i and ii (b) ii and iii (c) i and iii (d) i, ii and iii

126. Which of the following represents primary data?

- i. A scientist collects soil samples for analysis
- ii. Data compiled in a textbook
- iii. A business owner surveys customers directly

Which one is correct?

- (a) i and iii (b) i and ii (c) ii and iii (d) i, ii, and iii

127. Which of these are examples of secondary data?

- i. A report sourced from census data
- ii. A student conducting a direct experiment
- iii. Statistics extracted from a government database

Which one is correct?

- (a) i and iii (b) i and ii (c) ii and iii (d) i, ii, and iii

128. Which one is true of primary data?

- i. Original
- ii. Suitable
- iii. Reliable

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

129. Which statement is true about secondary data?

- i. Already published
- ii. Economical
- iii. Always up-to-date

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

130. Which one is true about secondary data?

- i. Easy to collect
- ii. Collected by someone else
- iii. Free from bias

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

131. Which is an advantage of primary data?

- i. Specific to the study
- ii. More reliable
- iii. Less time-consuming

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

3 Measures of Central Tendency

3.1 General Questions

132. Which statement is correct

- (a) Quartiles are well defined (b) Outliers affect Median
(c) Median is always present in data (d) Quadratic mean is widely used

133. Which measure is suitable for open-ended distribution?
 (a) Median (b) Mode (c) Geometric Mean (d) Arithmetic mean
134. Which is not a measure of central tendency?
 (a) Arithmetic mean (b) Mode (c) Range (d) Quadratic mean
135. When is the statement $AM = GM = HM$ true?
 (a) When the values are natural numbers (b) When all the values are equal
 (c) When all the values have equal frequency (d) When mode is greater than median
136. If a value is zero, which measure is not usable?
 (a) Arithmetic Mean (b) Harmonic Mean (c) Geometric Mean (d) Mode
137. How many measure of central tendency are there?
 (a) 2 (b) 3 (c) 4 (d) 5
138. Which measure of central tendency is suitable for qualitative variable?
 (a) Arithmetic Mean (b) Harmonic Mean (c) Quadratic Mean (d) Mode
139. In presence of negative values, which measure is not usable?
 (a) Arithmetic Mean (b) Geometric Mean (c) Quadratic Mean (d) Harmonic Mean
- Answer the next two questions based on the following information**

Accident	4	6	7	8	9
Frequency	2	0	4	5	1

140. Fifth Decile is –
 (a) 0 (b) 8.5 (c) 7.5 (d) 8
141. Which of the following is mode?
 (a) 4 (b) 8 (c) 0 (d) 7
142. Which measure always gives a value from within the values?
 (a) Arithmetic Mean (b) Geometric Mean (c) Median (d) Mode
143. Which one is not a proper measure of central tendency?
 (a) 2nd Quartile (b) Third Decile (c) 3rd Quintile (d) 110th Percentile
144. Which one is smallest?
 (a) $\sum_{i=1}^n (X_i - Median)^2$ (b) $\sum_{i=1}^n (X_i - \bar{X})^2$ (c) $\sum_{i=1}^n (X_i - \sigma)^2$ (d) $\sum_{i=1}^n (X_i - Mode)^2$
145. Which measure is not used in determining skewness?
 (a) Arithmetic Mean (b) Geometric Mean (c) Median (d) Mode
146. When is the relationship $AM = HM = GM$ true?
 (a) All values are equal (b) The values form a geometric progression
 (c) The values form an arithmetic progression (d) All values are distinct
147. In the presence of outlier(s), which measure of central tendency is suitable?
 (a) Arithmetic mean (b) Median (c) Quadratic mean (d) Power mean

148. Which measure is suitable for dealing with population growth?
 (a) Arithmetic Mean (b) Geometric Mean (c) Median (d) Harmonic mean
149. Which measure is best for calculating average rates of change over time?
 (a) Arithmetic Mean (b) Geometric Mean (c) Median (d) Harmonic Mean
150. Which measure is best for determining average income in a highly skewed income distribution?
 (a) Arithmetic Mean (b) Geometric Mean (c) Median (d) Harmonic Mean
151. Which can be measured from Ogive?
 (a) Arithmetic Mean (b) Geometric Mean (c) Median (d) Harmonic Mean
152. If a rate is defined as $R = \frac{c}{d}$, where c is constant, then which measure is perfect?
 (a) Weighted arithmetic mean (b) Harmonic mean
 (c) Quadratic mean (d) Weighted geometric mean
153. Which measure might have more than one value?
 (a) Arithmetic mean (b) Geometric mean (c) Quadratic mean (d) Mode
154. Which relationship is correct?
 (a) $AM \times GM = HM^2$ (b) $AM \times HM = GM^2$ (c) $AM \times HM = GM^3$ (d) $AM \div GM = HM^2$
155. The arithmetic mean and geometric mean of two non-zero positive numbers are 15 and 10, respectively. What is harmonic mean?
 (a) 6.61 (b) 6.67 (c) 7.66 (d) 6.76
156. For two non-zero positive numbers, the harmonic mean is 8 and the geometric mean is 12. What is the arithmetic mean?
 (a) 16 (b) 18 (c) 20 (d) 22
157. For two non-zero positive numbers, the harmonic mean is 10 and the arithmetic mean is 25. What is the geometric mean?
 (a) 7.07 (b) 20 (c) 25 (d) 30
158. The following are values of arithmetic, geometric, and harmonic mean, respectively
 i. {10, 12, 14}
 ii. {10, 10, 10}
 iii. {10, 8, 6}
 Which one is possible?
 (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii
 (a) {10, 12, 14} (b) {10, 09, 14} (c) {16, 16, 14} (d) {10, 12, 14}
159. Which one is a correct approximation of empirical relationship for a moderately skewed distribution?
 (a) $Mo = 3Me - 2\bar{X}$ (b) $Me = 3Mo - 2\bar{X}$ (c) $Mo = 2Me - 3\bar{X}$ (d) $Mo = Me - 2\bar{X}$
160. The values of mode and median of moderately skewed distribution are 60 and 65. What is value of the mean?
 (a) 70 (b) 67.5 (c) 72 (d) 55

161. For a moderately skewed distribution, the mean and median are 50 and 48 respectively. Find the mode.
 (a) 40 (b) 42 (c) 44 (d) 46
162. In a moderately skewed distribution, the mean and mode are 30 and 42 respectively. Find the median.
 (a) 30 (b) 34 (c) 38 (d) 42
163. For a moderately skewed distribution, if Mean = 50 and Median = 45, then the distribution is:
 (a) Symmetrical (b) Positively skewed (c) Negatively skewed (d) Bimodal
164. For a moderately skewed distribution, Mode = 75 and Median = 80. Find Mean.
 (a) 80.5 (b) 82.5 (c) 85.5 (d) 87.5
165. For a moderately skewed distribution, Mean = 60 and Mode = 48. Find Median.
 (a) 54 (b) 56 (c) 58 (d) 60
166. For a moderately skewed distribution, which of the following is true?
 (a) Mean = Mode = Median (b) Mean - Mode = 3(Mean - Median)
 (c) Mode = 3(Median - Mean) (d) Mean = 3(Median - Mode)

3.2 Arithmetic Mean

167. If $\bar{X} = 20$, what is the value of $\bar{Y}$, if $Y = 4X$?
 (a) 24 (b) 5
 (c) 80 (d) Cannot be determined
168. If $\bar{X} = 12$, what is $\bar{Y}$ when $Y = 3X$?
 (a) 4 (b) 12 (c) 24 (d) 36
169. If $\bar{X} = 30$, what is $\bar{Y}$ when $Y = 2X + 5$?
 (a) 35 (b) 60 (c) 65 (d) 70
170. If $Y = 4X - 8$ and $\bar{Y} = 20$, find $\bar{X}$.
 (a) 3 (b) 5 (c) 7 (d) 9
171. If each observation is multiplied by 3 and then increased by 4, the arithmetic mean of the original data will be:
 (a) Multiplied by 3 (b) Increased by 4
 (c) Multiplied by 3 and then increased by 4 (d) Unchanged
172. Let $Y = aX + b$, where a and b are constants. If $\bar{X} = 8$ and $\bar{Y} = 30$, and $a = 3$, $b = ?$.
 (a) 3 (b) 6 (c) 9 (d) 12
173. If $\bar{X} = 25$, what is $\bar{Y}$ when $Y = X - 10$?
 (a) 10 (b) 15 (c) 25 (d) 35
174. If each value of a variable is multiplied with 4, what would be the new arithmetic mean?
 (a) 4 times (b) twice (c) Thrice (d) Unchanged

175. If $\sum(x_i - k) = 0$, what is the value of k ?
 (a) n (b) $\bar{x}$ (c) x (d) $n\bar{x}$
176. If $\sum(x_i - a)^2$ is minimized, then the value of a is:
 (a) $\bar{x}$ (b) 0 (c) Median (d) Mode
177. Find the arithmetic mean: 6, 9, 12, $\dots$, 84
 (a) 40 (b) 45 (c) 50 (d) 55
178. The arithmetic mean of first 10 natural numbers is:
 (a) 6 (b) 8.5 (c) 5.5 (d) 5.6
179. Arithmetic Mean of first 25 natural numbers is –
 (a) 12 (b) 13 (c) 14 (d) 26
180. An equation is: $y = 5x + 9$. If $\bar{x} = 20$, $\bar{y} = ?$
 (a) 100 (b) 209 (c) 109 (d) 29
181. An equation is: $y = 5x + 9$. If $\bar{x} = 20$, what is $\bar{y}$?
 (a) 100 (b) 209 (c) 109 (d) 29
182. Given the relationship $y = 2x - 4$, and $\bar{x} = 15$, find the value of $\bar{y}$.
 (a) 26 (b) 34 (c) -26 (d) 35
183. Arithmetic Mean of two numbers is 25. If a number is 40, what is the other number?
 (a) 40 (b) 50 (c) 25 (d) 10
184. The Arithmetic Mean of two numbers is 30. If one number is 40, what is the other number?
 (a) 20 (b) 30 (c) 40 (d) 60
185. The Arithmetic Mean of two numbers is 35. If one number is 50, what is the other number?
 (a) 25 (b) 20 (c) 40 (d) 70
186. Number of students in two classes are 50 and 55 and their combined arithmetic mean (AM) of marks is 82. If AM of the first class is 75, what is the AM of the other class?
 (a) 88.36 (b) 88.40 (c) 84.55 (d) 78.33
187. The summation of deviation of each value from their arithmetic mean is –
 (a) 0 (b) 1 (c) 2 (d) 4
188. For grouped data, which formula is correct for Arithmetic Mean?
 (a) $\bar{X} = \frac{\sum f_i x_i}{\sum f_i}$ (b) $\bar{X} = \frac{\sum x_i}{N}$ (c) $\bar{X} = \frac{\sum f_i x_i}{n}$ (d) $\bar{X} = \frac{\sum f_i}{N}$
189. Arithmetic mean of the series 2, 12, 22, $\dots$, 92 is–
 (a) 45 (b) 46 (c) 47 (d) 55
190. What is the arithmetic mean of first n odd natural numbers?
 (a) $\frac{n+1}{n}$ (b) n (c) $n+1$ (d) $\frac{n+1}{2}$
191. What is the arithmetic mean of first n even natural numbers?
 (a) $\frac{n+1}{2}$ (b) $n+1$ (c) n (d) $\frac{n-1}{2}$

192. The arithmetic mean of first n natural numbers-

- (a) $\frac{n}{2}$ (b) $\frac{n+1}{2}$ (c) $\frac{n^2}{2}$ (d) $\frac{n^2-1}{2}$

193. Arithmetic means of three groups having equal no. of items are 30, 32, and 34. What is the combined mean?

- (a) 30.33 (b) 32.67 (c) 32.00 (d) 33.00

3.3 Harmonic Mean

194. Which formula is correct for harmonic mean?

- (a) $\frac{n}{\sum_{i=1}^n \frac{f_i}{x_i}}$ (b) $\frac{f_i}{\sum_{i=1}^n \frac{f_i}{x_i}}$ (c) $\frac{\sum f_i}{\sum_{i=1}^n \frac{f_i}{x_i}}$ (d) $\frac{\sum f_i}{\sum_{i=1}^n \frac{1}{x_i}}$

195. What is the harmonic mean of these values: 10, 12, 13, 15, 20, 25

- (a) 12.49 (b) 14.93 (c) 14.39 (d) 13.49

196. A rate is defined as $R = \frac{c}{d}$; c and d are arbitrary numbers. If c is constant, which mean is used?

- (a) Arithmetic Mean (b) Geometric Mean
(c) Harmonic Mean (d) Weighted Geometric Mean

197. A rate is defined as $R = \frac{c}{d}$; c and d are arbitrary numbers. If d is constant, which mean is used?

- (a) Arithmetic Mean (b) Geometric Mean
(c) Harmonic Mean (d) Weighted Geometric Mean
(a) Arithmetic Mean (b) Geometric Mean
(c) Harmonic Mean (d) Weighted Geometric Mean

198. Which is the representation of Harmonic Mean?

- (a) Mean of Reciprocal (b) Reciprocal of Mean
(c) Reciprocal of Mean of Reciprocal (d) None of the above

3.4 Geometric Mean

199. Which data set is suitable for Geometric Mean?

- (a) 1, -1, 2, 4, 6, 7 (b) 1, 2, 4, 8, 16, 32 (c) 0, 1, 2, 3, 4, 6 (d) 1, 1, 2, 3, 4, 4, 5

200. Find geometric mean: 2, 4, 8, 16

- (a) 6.65 (b) 6.56 (c) 5.66 (d) 5.56

Answer the next three questions based on the following information

The data collected in a research is this: 1, 2, 4, 8, 16, 32

201. Which measure is suitable?

- (a) Arithmetic Mean (b) Geometric Mean (c) Median (d) Mode

202. What is the arithmetic mean of the data?

- (a) 8.5 (b) 10 (c) 8 (d) 10.5

203. What is the geometric mean?

- (a) 8.5 (b) 5.66 (c) 6.55 (d) 16

3.5 Mode

204. Which of the following may be used to determine mode?
(a) Histogram (b) Frequency Curve (c) Ogive (d) Frequency Polygon
205. What is the mode the set: 7, 8, 8, 9, 9, 13, 17, 9, 8, 8
(a) 17 (b) 9
(c) 8 (d) Cannot be determined
206. What is the mode of the data set: 4, 7, 2, 4, 9, 4, 2, 9?
(a) 2 (b) 4 (c) 9 (d) 7
207. Which of the following best defines the mode of a data set?
(a) The middle value when data are arranged in order
(b) The average of all the values
(c) The value that occurs most frequently
(d) The difference between highest and lowest values
208. Find the mode of the following frequency distribution:

Value	2	3	4	5	6
Frequency	3	5	2	7	1

- (a) 3 (b) 5 (c) 6 (d) 5
209. In a symmetrical unimodal distribution, which of the following is usually true?
(a) Mean < Median < Mode (b) Mean > Median > Mode
(c) Mean = Median = Mode (d) Mode > Mean

3.6 Median

210. Which can be measured from the Ogive?
(a) Arithmetic Mean (b) Geometric Mean (c) Median (d) Mode
211. Median can be determined from the—
(a) Histogram (b) Frequency curve (c) Ogive (d) Pie Chart

3.7 Partition Values

3.8 Situation Set

Answer the next three questions based on the following information

The following table shows weekly production of milk (in liters) by different varieties of cows.

Interval	10-20	20-30	30-40	40-50	50-60	60-70
Frequency	5	12	18	25	20	10

212. What is the median?

- (a) 43 (b) 44 (c) 45 (d) 50

213. What is the lower limit of class interval for first quartile?

- (a) 10 (b) 20 (c) 30 (d) 40

214. What is the 3rd quartile?

- (a) 55.75 (b) 43.75 (c) 53.15 (d) 53.75

Answer the next two (2) questions based on the following information

Class	≤ 20	20-25	25-50	50-60	69-70	≥ 70
Frequency	5	10	10	7	5	3
Cumulative Frequency	5	15	25	32	37	40

215. How many values are between 20 and 70?

- (a) 20 (b) 32 (c) 35 (d) 37

216. Which one is the median class?

- (a) 20-25 (b) 25-50 (c) 50-60 (d) 60-70

217. What is the median of the following values: 4, 5, 2, 1, 8, 3

- (a) 1.5 (b) 2 (c) 3.5 (d) 4

Answer the next three questions as per the following information.

42 44 59 64 70 72 74 91 94 are 9 values.

218. What is the 50th percentile?

- (a) 64 (b) 70 (c) 72 (d) 71

219. Below which value lie 70 percent values?

- (a) 42 (b) 44 (c) 59 (d) 74

220. Above which value lie 30% observations?

- (a) 3rd Quartile (b) Median (c) 30th Percentile (d) 70th percentile

Answer the next three questions as per the following information.

42 44 59 64 70 72 74 91 94 are 9 values.

221. What is the median?

- (a) 64 (b) 70 (c) 72 (d) 71

222. What is the first quartile?

- (a) 42.4 (b) 44.7 (c) 51.5 (d) 64.2

223. Above which value lie 60% observations?

- (a) 70.4 (b) 72.0 (c) 74.6 (d) 66.4

3.9 Multiple Completion

224. **Inappropriate for algebraic analysis–**

- i. Median
- ii. Mode
- iii. Geometric Mean

Which one is true?

- (a) i (b) ii (c) i & ii (d) ii & iii

225. **With negative observations, which cannot be used**

- i. Arithmetic Mean
- ii. Geometric Mean
- iii. Harmonic Mean

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

226. **A good measure of central tendency -**

- i. is loosely defined
- ii. takes into consideration all values
- iii. easily understandable

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

227. **A good measure of central tendency -**

- i. is not affected by extreme values
- ii. represents the entire dataset accurately
- iii. is difficult to compute

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

228. **A good measure of central tendency -**

- i. is stable for different samples
- ii. provides a single representative value
- iii. ignores extreme values completely

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

229. **Median is –**

- i. Affected by extreme values
- ii. Rigidly defined
- iii. Suitable for open-ended distributions

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

230. **Mode is –**

- i. The most frequently occurring value
- ii. Unaffected by extreme values
- iii. Always unique in a dataset

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

231. A rate is defined as $R = \frac{c}{d}$; c and d are arbitrary numbers. If neither c or d is constant, which mean is used?

- i. Weighted Arithmetic Mean
- ii. Weighted Harmonic Mean
- iii. Harmonic Mean

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

232. What is true of harmonic mean?

- i. uses all values in the data
- ii. undefined if the any value is zero
- iii. affected by extreme values

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

233. Arithmetic Mean is –

- i. Rigidly defined
- ii. Unaffected by sample fluctuation
- iii. Suitable for algebraic analysis

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

234. Which of the following are true about the median?

- i. It is the middle value when data is ordered
- ii. It is unaffected by extreme outliers
- iii. It is always equal to the mean in a symmetrical distribution

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

235. Which of the following are true about the arithmetic mean?

- i. It is affected by extreme values
- ii. The sum of deviations from the mean is zero
- iii. It can be calculated for open-ended distributions

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

236. Which of the following are true about the mode?

- i. A distribution can have more than one mode
- ii. It is the most frequently occurring value
- iii. It is always unique

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

4 Measures of Dispersion

237. Which measure of dispersion is expressed in the same units as the original data?

- (a) Variance (b) Standard deviation
(c) Coefficient of variation (d) Mean deviation about mean

238. If $Q_1 = 25$ and $QD = 10$, what is the value of Q_3 ?
- (a) 45 (b) 70 (c) 10 (d) 25
239. What is the variance of a constant c ?
- (a) 1 (b) 0 (c) -1 (d) ∞
240. What is the correct formula for the variance of a sample?
- (a) $\frac{\sum x_i^2}{n} - \frac{\bar{x}}{n}$ (b) $\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2$ (c) $\sum x^2$ (d) $\sum 2^x$
241. Two sets of data are $X = 2, 4, 6$ and $Y = 4, 6, 8$; which has higher variance?
- (a) X
(b) Y
(c) $V(X) = V(Y)$
(d) Cannot be determined from the given information
242. Which set of data looks more scattered?
- (a) -1, 1, 2 (b) -5, 6, 7 (c) -10, 10, 30 (d) 5, 8, 20
243. Which of the following is the best measure of dispersion?
- (a) Range (b) Mean deviation
(c) Standard deviation (d) Coefficient of variation
244. What is the minimum possible value of standard deviation?
- (a) ∞ (b) -1 (c) 0 (d) 1
245. For two values, the range is found to be 8. What are the values of mean deviation and standard deviation, respectively?
- (a) (2,4) (b) (4,4) (c) (4,8) (d) (8,8)
246. If the standard deviation of a dataset is 4, what is its variance?
- (a) 2 (b) 4 (c) 8 (d) 16
247. If each observation is multiplied by 3, what happens to the standard deviation?
- (a) Becomes one-third (b) Becomes 3 times (c) Remains the same (d) Becomes 9 times
248. Standard deviation is always:
- (a) Positive (b) Zero (c) Non-negative (d) Negative
249. The coefficient of variation is calculated as:
- (a) $\frac{\text{Mean}}{\text{SD}} \times 100$ (b) $\frac{\text{SD}}{\text{Mean}} \times 100$ (c) $\frac{\text{Mean}}{\text{MD}} \times 100$ (d) $\frac{\text{MD}}{\text{Mean}} \times 100$
250. If the mean of a dataset is 50 and standard deviation is 10, the coefficient of variation is:
- (a) 10% (b) 15% (c) 20% (d) 25%
251. Coefficient of variation is used to compare:
- (a) Datasets with same unit (b) Datasets with different units or means
(c) Only variances (d) Only means
252. Mean deviation is minimized when taken from:
- (a) Mean (b) Median (c) Mode (d) Geometric mean

253. If $Q_1 = 25$ and $Q_3 = 45$, the quartile deviation is:
 (a) 5 (b) 7.5 (c) 10 (d) 12.5
254. Quartile deviation ignores:
 (a) The upper half of data (b) The lower half of data
 (c) The middle 50% of data (d) The extreme 25% in both tails
255. Which measure of dispersion is most affected by outliers?
 (a) Standard deviation (b) Range (c) Quartile deviation (d) Mean deviation
256. For two values, the range is found to be 12. What are the values of mean deviation and standard deviation
 (a) (2,4) (b) (4,4) (c) (6, 6) (d) (8,8)
257. The mean and coefficient of variation of a distribution are 5 and 30%, respectively. What is the value of standard deviation?
 (a) 1.5 (b) 6.5 (c) 7.6 (d) 10.2
258. What is the correct formula for the standard deviation?
 (a) $\sqrt{\frac{1}{n} \sum (x_i - \bar{x})^2}$ (b) $\sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2}$ (c) $\frac{1}{n} \sum (x_i - \bar{x})$ (d) $\sqrt{\sum (x_i - \bar{x})}$
259. What is the standard deviation of the first n natural numbers?
 (a) $\sqrt{\frac{(n^2-1)}{6}}$ (b) $\sqrt{\frac{(n^2-1)}{12}}$ (c) $\sqrt{\frac{n(n+1)(2n+1)}{6n}}$ (d) $\sqrt{\frac{n(n+1)}{2}}$
260. The Mean Deviation of two unequal numbers is 3. What is their range?
 (a) 1.5 (b) 3 (c) 6 (d) 12
261. What is the standard deviation of first 10 natural numbers?
 (a) 2.87 (b) 3.02 (c) 0 (d) 2.78
262. Which measure is unit-free?
 (a) Range (b) Mean deviation
 (c) Standard deviation (d) Coefficient of variation
263. Which measure is suitable for an open-ended distribution?
 (a) Range (b) Mean deviation (c) Standard deviation (d) Quartile deviation

4.1 Situation Set

Answer the next two questions based on the following information.

The temperatures (in $^{\circ}C$ of two cities in a country are 30 and 35.

264. What is their Mean deviation?
 (a) 1.2 (b) 2.5 (c) 3.0 (d) 5.5
265. What is the coefficient of variation?
 (a) 2.7% (b) 8.3% (c) 5.8% (d) 7.7%

4.2 Multiple Completion

266. Which of the following does not depend on change of origin?

- i. Median
- ii. Standard deviation
- iii. Regression coefficient

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

267. Which of the following are true about the range?

- i. It is the difference between the maximum and minimum values
- ii. It is affected by extreme outliers
- iii. It considers all observations in the dataset

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

268. Which of the following are true about variance?

- i. It is always non-negative
- ii. It is measured in squared units of the original data
- iii. It becomes zero when all observations are equal

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

269. Which of the following are true about standard deviation?

- i. It has the same unit as the original data
- ii. It is the square root of variance
- iii. A larger standard deviation indicates greater dispersion

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

270. Which of the following are true about the coefficient of variation?

- i. It is a unit-free measure of relative dispersion
- ii. It is defined as $\frac{\sigma}{\mu} \times 100\%$
- iii. It is useful for comparing variability across different datasets

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

271. Which of the following are true about the interquartile range?

- i. It is the difference between the third and first quartiles
- ii. It is less affected by outliers than the range
- iii. It measures the spread of the middle 50

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

272. Which of the following are true about mean deviation?

- i. It is the average of absolute deviations from a central value
- ii. It ignores signs of deviations
- iii. It is always zero for symmetrical distributions

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

273. Dispersion has application in measuring —

- i. regression
- ii. correlation
- iii. range

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

274. Standard deviation —

- i. depends on all values
- ii. is not affected by outliers
- iii. can be analyzed algebraically

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

5 Moments, Skewness, and Kurtosis

5.1 Moments

275. What is the purpose of moments?

- (a) To find the center of data
- (b) To find dispersion
- (c) To find relationship between skewness and kurtosis
- (d) To uniquely identify a distribution

276. The moments around the origin are called —

- (a) Central moments (b) Raw moments
(c) First raw moment (d) Measures of dispersion

277. The Range of Karl Pearson's measure of skewness —

- (a) (0, 1) (b) (-1, 1) (c) (-3, 3) (d) (0, ∞)

278. Which is not a type of Moments

- (a) Central Moments (b) Raw Moments (c) Corrected Moments (d) Rectified Moments

279. The second moment around w is —

- (a) $\frac{\sum (x_i - \bar{x})^n}{w}$ (b) $\frac{\sum (x_i - \bar{x})^2}{w}$ (c) $\frac{\sum (x_i - w)^2}{n}$ (d) $\frac{\sum (x_i - w)^n}{2}$

280. Which relationship is correct?

- (a) $\mu'_1 = \bar{x} + a$ (b) $\mu'_1 = \bar{x} - a$ (c) $\mu'_2 = \bar{x} + a$ (d) $\mu_1 = \bar{x} - a$

281. What is formula of rth raw moment for grouped data about a?

- (a) $\frac{\sum f_i (x_i - a)^r}{n}$ (b) $\frac{\sum f_i (x_i - \bar{x})^r}{n}$ (c) $\frac{\sum (x_i - a)^r}{n}$ (d) $\frac{\sum (x_i + a)^r}{n}$

282. Which quantity uniquely characterizes a distribution?

- (a) Median (b) Quantile (c) Moments (d) Trend

Which one is correct?

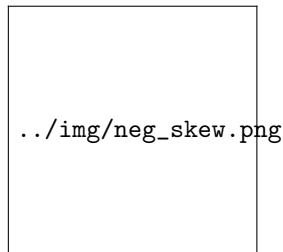
- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

283. Which can be used to measure dispersion?
 (a) μ'_2 (b) μ_1 (c) μ_2 (d) μ'_1
284. The formula of coefficient of variance (CV) is –
 (a) $\frac{\sqrt{\mu_2}}{n} \times 100$ (b) $\frac{\mu_2}{\mu_1} \times 100$ (c) $\frac{\sqrt{\mu_2}}{\bar{x}} \times 100$ (d) $\frac{\mu_3}{\sigma} \times 100$
285. First moment around zero is –
 (a) 0 (b) 1 (c) -1 (d) Arithmetic Mean
286. Which moment is equal to zero?
 (a) First raw moment around 1 (b) Second central moment
 (c) First central moment (d) Second raw moment around 0
287. Which might have a negative value?
 (a) μ_4 (b) μ_3 (c) μ'_2 (d) μ_2
288. 2nd Central Moment is –
 (a) $\mu_2 - \mu'_1$ (b) $\mu_2 + \mu'_1$ (c) $\mu_2 - \mu'^2_1$ (d) $\mu'_2 - \mu'^2_1$
289. First central moment is equal to –
 (a) 1 (b) 0 (c) -1 (d) $\bar{x} - a$
290. First moment around a is equal to –
 (a) 1 (b) 0 (c) -1 (d) $\bar{x} - a$
291. The first raw moment about 3 is -5. What is the value of arithmetic mean?
 (a) 2 (b) -2 (c) 0 (d) 8
292. The first raw moment about 4 is -4. What is the value of arithmetic mean?
 (a) 2 (b) -2 (c) 0 (d) 8
293. The first raw moment about 0 is 2. What is the value of arithmetic mean?
 (a) 2 (b) -2 (c) 0 (d) 8
294. The arithmetic mean of a variable is 4. What is the first raw moment around 2?
 (a) 2 (b) -2 (c) 0 (d) 8
295. The arithmetic mean of a variable is 10. What is the first raw moment around 0?
 (a) 10 (b) -2 (c) 0 (d) 8
296. The arithmetic mean of a variable is 2.6. What is the first raw moment around 6?
 (a) 2.2 (b) -3.4 (c) 0.1 (d) 1.8
297. If the values in a dataset have mean 4.8, what is the first moment about the mean?
 (a) 0 (b) 4.8 (c) 1.0 (d) -4.8
298. The mean of a variable is 3.2. Find the first raw moment around 0.
 (a) 3.2 (b) -3.2 (c) 0 (d) 1.2
299. The first raw moment around 0 of a data set is 5. What is the arithmetic mean of the data?
 (a) 3 (b) 4 (c) 5 (d) 6

300. The first raw moment around 5 of a data set is 15. What is the arithmetic mean of the data?
 (a) 8 (b) 20 (c) 12 (d) 15
301. The first raw moment around 3 of a data set is 18. What is the arithmetic mean of the data?
 (a) 6 (b) 17 (c) 28 (d) 21
302. The first raw moment around 10 of a data set is 50. What is the arithmetic mean of the data?
 (a) 52 (b) 24 (c) 60 (d) 40
303. Moments can be—
 i. positive
 ii. not negative
 iii. positive or negative
 Which one is correct?
 (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii
304. The second central moment of the natural numbers 1 through 12 is —
 (a) 12.93 (b) 11.92 (c) 10.94 (d) 12.60
305. The second raw moment around zero of the natural numbers 5 through 15 is —
 (a) 120 (b) 115 (c) 100 (d) 110

5.2 Skewness

306. What kind of distribution is clustered around the mean?
 (a) Left-skewed (b) Symmetrical (c) Right-skewed (d) None of the above
307. The following graph is an example of —



- (a) Positive Skew (b) Negative Skew (c) No Skew (d) Not detectable
308. For a symmetrical distribution, what is the value of β_1 ?
 (a) 0 (b) 1 (c) -1 (d) ∞

Answer the next SIX questions based on the following information

../../../../slide/img/skewness_clean.jpg

309. The curve (a) is an example of
 (a) Positive Skew (b) Negative Skew (c) No Skew (d) Not detectable
310. The curve (b) is an example of
 (a) Positive Skew (b) Negative Skew (c) No Skew (d) Not detectable
311. In Image (b), what is denoted by 4th value?
 (a) Mean (b) Median (c) Mode (d) All of the above
312. In Image (c), what is in 6th value?
 (a) Mean (b) Median (c) Mode (d) None of the above
313. What is the value corresponding to the position 3?
 (a) Mean (b) Median (c) Mode (d) None of the above
314. What is the value corresponding to the position 7?
 (a) Mean (b) Median (c) Mode (d) None of the above
315. If $\gamma_1 > 0$, the data is -
 (a) Negatively skewed (b) Positively skewed (c) Symmetric (d) Uncertain
316. Which relationship is correct?
 (a) $M_o = 2Me - \bar{x}$ (b) $M_o = 3Me - \bar{x}$ (c) $M_o = 3Me - 2\bar{x}$ (d) $M_o = 2Me - 3\bar{x}$
317. Characteristics of a skewed distributon are –
 i. $Mean \neq Median \neq Mode$
 ii. Differences of upper and lower quartiles from median are unequal
 iii. Frequency curve is asymmetric
318. In a distribution, $\mu_2 = 25$, $\mu_3 = 20$, and $\mu_4 = 2200$; the distribution is –
 (a) Negatelky skewed (b) leptokurtic (c) Platykurtic (d) Symmetric

319. For a data, $Q_3 = 41.6, Q_1 = 17.2, \text{Median} = 29, \&AM = 30$; What is Coefficient of skewness?
 (a) 24.4 (b) 1 (c) 0.03 (d) 29.45
320. In case of positive skewness, which one is correct?
 (a) $\text{Mean} > \text{Median} > \text{Mode}$ (b) $\text{Mean} < \text{Median} < \text{Mode}$
 (c) $\text{Mean} = \text{Median} = \text{Mode}$ (d) $\text{Mean} > \text{Median} < \text{Mode}$
321. For a symmetrical distribution, $\beta_1 =$ —
 (a) 1 (b) -1 (c) 0 (d) 3
322. $\sqrt{\beta_1} = -0.23$ implies—
 (a) Left Skew (b) Symmetry (c) Right Skew (d) Mesokurtic
323. $\gamma_1 = 0.43$ implies—
 (a) Left Skew (b) Symmetry (c) Right Skew (d) Mesokurtic
324. $\gamma_1 = 0.0001$ implies—
 (a) Left Skew (b) Symmetry (c) Right Skew (d) Mesokurtic
325. First 3 moments about 2 are 1, 2 and 8, respectively. What is the arithmetic mena?
 (a) 1 (b) 2 (c) 3 (d) 4
326. What is the second central moments of first 10 natural numbers?
 (a) 9.90 (b) 9.09 (c) 8.25 (d) 5.67
327. Frequencies of low and high values are smaller in – distribution
 (a) Positively skewed (b) Negatively skewed (c) Symmetric (d) Mesokurtic
328. Frequencies of higher values are smaller and of low values are higher in – distribution
 (a) Positively skewed (b) Negatively skewed (c) Symmetric (d) Mesokurtic
329. Frequencies of higher values are higher and of low values are lower in – distribution
 (a) Positively skewed (b) Negatively skewed (c) Symmetric (d) Mesokurtic
330. In a postively-skewed distribution—
 i. Frequencies of higher values are lower
 ii. Frequencies of low values are higher
 iii. Frequencies of higher values are higher
 Which one is correct?
 (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii
331. In a negatively-skewed distribution—
 i. Frequencies of higher values are lower
 ii. Frequencies of low values are lower
 iii. Frequencies of higher values are higher
 Which one is correct?
 (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

332. **In a symmetric distribution—**

- i. Frequencies of higher values are lower
- ii. Frequencies of low values are higher
- iii. Frequencies of low values are lower

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

333. **Which formula is correct for determining skewness?**

- (a) $\gamma_1 = \sqrt{\frac{\mu_3^2}{\mu_2^3}}$ (b) $\gamma_1 = \sqrt{\beta_1^2}$ (c) $\gamma_1 = \sqrt{\frac{\mu_3}{\mu_2^2}}$ (d) $\frac{\mu_2}{\sqrt{\mu_3^2}}$

5.2.1 Situation Set

Answer the next THREE questions based on the following information

The arithmetic mean, Pearson's coefficient of skewness, and coefficient of variation of a distribution are 50, -0.4, and 40 %, respectively.

334. **What is the standard deviation?**

- (a) 50 (b) 20 (c) 30 (d) 40

335. **The Mode is –**

- (a) 45 (b) 50 (c) 60 (d) 58

336. **The Median is**

- (a) 52.67 (b) 5 (c) 20 (d) 10.39

Answer the next two questions based on the following information

The first and second moments of a distribution around 2 are 13 and 195, respectively.

337. **What is the arithmetic mean?**

- (a) 15 (b) 12 (c) 11 (d) 26

338. **What is its variance?**

- (a) 20 (b) 21 (c) 26 (d) 25

5.2.2 Multiple Completion

339. **In a negatively skewed distribution –**

- i. there is a long tail on the left
- ii. median is greater than mean
- iii. high values have frequencies

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

5.3 Kurtosis

340. Which curve is platykurtic?

../img/kurtosis.jpg

- (a) A (b) B (c) C (d) None

341. How many types of kurtosis are there?

- (a) 2 (b) 3 (c) 4 (d) 5

342. The standard deviation of a mesokurtik distribution is 2. What is the value of the 4th central moment?

- (a) 4 (b) 8 (c) 16 (d) 48

343. $\beta_2 = \sqrt{9}$ implies data are—

- (a) Leptokurtic (b) Platykurtic (c) Mesokurtic (d) Symmetric

344. $\beta_2 = 4$ implies data are—

- (a) Leptokurtic (b) Platykurtic (c) Mesokurtic (d) Symmetric

345. $\beta_2 = 3$ implies data are—

- (a) Leptokurtic (b) Platykurtic (c) Mesokurtic (d) Symmetric

346. $\beta_2 = 1$ implies data are—

- (a) Leptokurtic (b) Platykurtic (c) Mesokurtic (d) Symmetric

347. The relationship between β_2 and γ_2 is —

- (a) $\beta_2 = \gamma_2 - 3$ (b) $\gamma_2 = \beta_2 - 3$ (c) $\gamma_2 = 3\beta_2$ (d) $\gamma_2 = \frac{\beta_2}{3}$

348. For a mesokurtik distribution, $\beta_2 =$ —

- (a) 0 (b) -3 (c) 3 (d) 1

349. What is the relationship between γ_2 and β_2 ?

- (a) $\gamma_2 = \beta_2 + 3$ (b) $\gamma_2 = 2\beta_2 - 3$ (c) $\gamma_2 = \beta_2 - 1$ (d) $\gamma_2 = \beta_2 - 3$

5.4 Misc

350. What is formula of the left inner fence for a box and whisker plot?

- (a) $Q_1 - 1.5 \times IQR$ (b) $Q_3 + 1.5 \times IQR$ (c) $Q_1 - 3 \times IQR$ (d) $Q_3 + 1.5 \times IQR$

351. What is the formula of IQR?

- (a) $IQR = Q_3 + Q_1$ (b) $IQR = Q_3 - Q_1$ (c) $IQR = 2Q_3 - Q_1$ (d) $IQR = \frac{Q_3 - Q_1}{2}$

352. Which is not used in constructing Box & Whisker Plot?

- (a) Mode (b) X_L (c) Q_1 & Q_3 (d) Q_1, Q_2 & Q_3

353. In a symmmatric distribution—

i. Arithmetic Mean = Mode = Median

ii. $Q_2 - Q_1 = Q_3 - Q_2$

iii. $Q_1 - X_L = X_H - Q_3$

Which one is true?

- (a) i & ii (b) ii & iii (c) i & iii (d) i, ii & iii

5.5 Box and Whisker Plot

354. The following values represent the quartiles of a data set:

- $Q_1 = 25$
- $Q_2 = 50$
- $Q_3 = 75$

What is the interquartile range (IQR)?

- (a) 25 (b) 50 (c) 75 (d) 100

355. In a box and whisker plot, the following statements hold true:

- i. The length of the box represents the interquartile range (IQR).
- ii. The whiskers extend from the minimum to the maximum data values.
- iii. The median is represented by the top of the box.

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

5.6 Five Number Summary

356. In a given data set, the following values are recorded:

- i. The interquartile range is the difference between Q_3 and Q_1 .
- ii. The median is always equal to the mean.
- iii. The maximum value in the five-number summary is the largest data point.

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

357. The five-number summary of a data set consists of the following:

- i. Minimum value
- ii. Maximum value
- iii. Median (Q_2), First Quartile (Q_1), Third Quartile (Q_3)

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

358. Which is not included in five number summary?

- (a) Arithmetic Mean (b) X_H (c) Q_2 (d) Q_3

5.7 Situation Set

Answer the next THREE questions based on the following information

For a frequency distribution, the following central moments are given:

$$\mu_2 = 4, \quad \mu_3 = -4, \quad \mu_4 = 48$$

359. What is the variance of the distribution?

- (a) 2 (b) 4 (c) 8 (d) 16

360. What is Karl Pearson's coefficient of skewness based on moments ($\sqrt{\beta_1}$)?

- (a) -1.0 (b) -0.5 (c) 0.5 (d) 1.0

361. What is the kurtosis (β_2) of the distribution?

- (a) 2.5 (b) 3.0 (c) 3.5 (d) 4.0

6 Correlation and Regression

6.1 Correlation

362. The correlation coefficient between two variables lies between:

- (a) 0 and 1 (b) -1 and 0 (c) -1 and +1 (d) $-\infty$ and $+\infty$

363. The correlation coefficient is unaffected by:

- (a) Change of origin only (b) Change of scale only
(c) Both change of origin and scale (d) Neither

364. A correlation coefficient of -0.85 indicates:

- (a) Weak positive correlation (b) Strong positive correlation
(c) Weak negative correlation (d) Strong negative correlation

365. If the correlation coefficient between X and Y is 0.6, the correlation between 2X and 3Y is:

- (a) 0.6 (b) 0.2
(c) 1.8 (d) Cannot be determined

366. Correlation coefficient is the geometric mean of:

- (a) Two regression coefficients (b) Two variances
(c) Two means (d) Two standard deviations

367. When all points lie exactly on a straight line with positive slope, the correlation coefficient is:

- (a) 1 (b) 0 (c) -1 (d) Infinity

368. If X and Y are independent, the correlation coefficient is:

- (a) 1 (b) -1 (c) 0 (d) Depends on sample

369. **The rank correlation coefficient is used when:**
- (a) Data is continuous (b) Variables are quantitative
(c) Data is in ranks or ordinal (d) Sample size is large
370. **Given covariance = 15, SD of X = 5, SD of Y = 4, the correlation coefficient is:**
- (a) 0.60 (b) 0.75 (c) 0.80 (d) 1.20
371. **The value of Spearman's rank correlation ranges between:**
- (a) -1 and +1 (b) 0 and 1 (c) -1 and 0 (d) $-\infty$ and $+\infty$
372. **If the regression coefficients are 0.8 and 0.5, the correlation coefficient is:**
- (a) 0.40 (b) 0.63 (c) 0.80 (d) 1.30
373. **If the correlation coefficient is zero, the variables are:**
- (a) Perfectly positively correlated (b) Perfectly negatively correlated
(c) Uncorrelated (d) Highly correlated
374. **Who proposed the formula of correlation coefficient?**
- (a) R. A. Fisher (b) Bowley (c) Spearman (d) Karl Pearson
375. **Which of the following has the strongest correlation?**
- (a) -0.92 (b) 0.67 (c) 0.50 (d) 0.91
376. **The lowest possible value of the correlation coefficient —**
- (a) 1 (b) 0 (c) $-\infty$ (d) -1
377. **The linear association between two random variables is called —**
- (a) Correlation (b) Regression (c) Randomness (d) Regularity
378. **Which measures the strength of linear association between two random variables?**
- (a) Correlation (b) Regression (c) Correlation coefficient (d) Regression coefficient
379. **Karl Pearson's method of determining the strength of correlation is not applicable for —**
- (a) Qualitative variable (b) Quantitative variable (c) Discrete variable (d) Continuous variable
380. **For two independent variables, the value of the correlation coefficient is —**
- (a) -1 (b) 1 (c) ∞ (d) 0
381. **Two variables having changes in same direction at same rates display —**
- (a) Perfect negative correlation (b) Partial positive correlation
(c) Perfect positive correlation (d) Partial negative correlation
382. **Two variables having changes in opposite direction at same rates display —**
- (a) Perfect negative correlation (b) Partial positive correlation
(c) Perfect positive correlation (d) Partial negative correlation
383. **Two variables having changes in same direction at different rates display —**
- (a) Perfect negative correlation (b) Partial positive correlation
(c) Perfect positive correlation (d) Partial negative correlation

384. **Two variables having changes in opposite direction at different rates display —**
 (a) Perfect negative correlation (b) Partial positive correlation
 (c) Perfect positive correlation (d) Partial negative correlation
385. **When a variable is changing, but another is not affected, it is called**
 (a) Perfect negative correlation (b) Partial positive correlation
 (c) Perfect positive correlation (d) Zero correlation
386. **TEXT**
 (a) Choice (b) Choice (c) Choice (d) Choice
387. **Which of the following coefficients indicates strong negative correlation?**
 (a) -0.5 (b) -2 (c) 0.67 (d) -0.94
388. **Which of the following has the strongest correlation?**
 (a) -0.5 (b) -2 (c) 0.67 (d) -0.94
389. **If $b_1 = 0.25$ and $b_2 = -0.43$, which implies higher impact of the independent variable?**
 (a) b_1 (b) b_2 (c) Cannot be determined (d) Equal impact

6.2 Situation Set

Answer the next two questions based on the following information

A study was conducted to find the impact of study hour on students' GPA and the following was found:

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = 30, \sum (x_i - \bar{x})^2 = 45, \text{ and } \sum (y_i - \bar{y})^2 = 55$$

390. **What is the value of correlation coefficient?**
 (a) 0.50 (b) 0.60 (c) -0.60 (d) -0.50
391. **What is the value of b_{yx} ?**
 (a) 0.58 (b) -0.67 (c) 0.67 (d) -1.75

6.3 Multiple Completion

392. **Which of the following are true about correlation?**

- i. It measures the strength of linear relationship
- ii. It is unitless
- iii. Correlation implies causation

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

7 Time Series

393. Which is not a time series data?

- (a) Number of calls received per week (b) No. of road accidents on different days
(c) No. of earthquakes in different regions (d) No. of particles decayed in each second

394. Which is not a time series data?

- (a) Daily closing prices of a stock (b) Annual temperature records of a city
(c) Number of students in a each class (d) Number of visitors to a website each day

395. Which is an example of time series data?

- (a) Number of calls received by a call center each month
(b) Height of children at different ages
(c) Total salary of all employees at a company
(d) Population of different countries in 2020

396. Which is a type of trend?

- i. Linear trend
ii. Non-linear trend
iii. Cyclic trend

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

397. Which can measure trend most precisely?

- (a) Graphical method (b) Semi-average method
(c) Moving average method (d) Quarter-average method

398. Which is the multiplicative time series model?

- (a) $Y_t = T_t \times S_t \times C_t \times R_t$ (b) $Y_t = T_t \times D_t \times C_t \times R_t$
(c) $Y_t = T_t \times P_t \times C_t \times R_t$ (d) $Y_t = T_t \times G_t \times C_t \times R_t$

399. In additive model, in the long run, $\sum R_t =$ --

- (a) 0 (b) 1 (c) Undefined (d) Infinity

400. In multiplicative time series model, in the long run, $\sum R_t =$ --

- (a) 0 (b) 1 (c) Undefined (d) Infinity

Answer the next two questions based on the following information

Commodity wise export shipments (In million US\$) of Frozen and live fish in Bangladesh are given below.

Months	2022-23 (July-Dec)	2023-24 (Jan-Jun)	2022-23 (July-Dec)
Amount	246.38	175.19	215.13

Table 1: Source:BB

401. Which component of time series is most evident?

- (a) Irregular variation (b) Cyclic variation (c) Trend (d) Seasonal variation

402. Which value is most probable in the next period?
 (a) 200 (b) 190 (c) 130 (d) 220
403. A linear trend goes along a –
 (a) a curved line (b) a wave (c) straight line (d) circle
404. Which of the following is an example of seasonal variation in a time series?
 (a) Increase in ice cream sales during summer (b) Rising fuel prices over decades
 (c) Stock market crash (d) Unemployment rate changes due to war
405. Which business is most likely to experience strong seasonal variation in its sales?
 (a) A supermarket (b) A toy store (c) A furniture store (d) A gas station
406. Which of the following is an example of cyclic variation in a time series?
 (a) Boom and recession phases in an economy
 (b) Increase in electricity consumption during summer
 (c) High demand for umbrellas during the rainy season
 (d) Sudden decline in stock prices due to a pandemic
407. Which of the following is an example of a trend in a time series?
 (a) Gradual increase in global average temperature
 (b) Increase in ice cream sales during summer
 (c) Fluctuations in stock prices due to news events
 (d) Sudden drop in airline bookings due to a storm
408. Which type of trend is usually observed in a country's population growth over several decades?
 (a) Upward trend (b) Downward trend (c) Seasonal trend (d) Cyclic trend
409. Which of the following best represents a downward trend in a time series?
 (a) Declining birth rates in a country over several decades
 (b) Increase in online shopping during holiday seasons
 (c) Fluctuations in stock market prices
 (d) Sudden rise in fuel prices due to a crisis
410. Which factor is most likely to contribute to an upward trend in a company's annual revenue?
 (a) Improved marketing strategies over time (b) Seasonal discounts and promotions
 (c) Short-term fluctuations in customer demand (d) Unpredictable supply chain disruptions
411. Which factor is most likely to cause cyclic variation in a time series?
 (a) Festive shopping trends (b) Long-term business cycles
 (c) Daily fluctuations in temperature (d) Random fluctuations in demand
412. A non-linear trend goes along a –
 (a) a curved line (b) a wave (c) a cubic pattern (d) Any of the above
413. Which measure of trend is subjective?
 (a) Semi-average method (b) Graphical method
 (c) Moving average method (d) None of the above

Answer the next THREE questions based on the following information

Year	2016	2017	2018	2019	2020	2021	2022	2023
USD Exchange Rate	78.35	79.49	82.87	83.26	84.60	84.37	85.80	106.70

Table 2: Source–Investing.com

414. What is the second value of semi-average method?

- (a) 85.40 (b) 90.37 (c) 91.73 (d) 89.78

415. What kind of a trend do the data have?

- (a) Upward (b) Downward
(c) Both upward & downward (d) No trend

416. Which component of time series is visible in the later part of the data?

- (a) Seasonal Variation (b) General Trend (c) Irregular Variation (d) Cyclic Variation

Answer the next THREE questions based on the following information

Year	2015	2016	2017	2018	2019	2020	2021	2022
Average Temperature (°C)	22.5	23.0	24.2	24.5	25.0	25.5	26.0	27.0

Table 3: Source–National Weather Service

417. What is the second value of the semi-average method?

- (a) 25.75 (b) 26.00 (c) 25.88 (d) 24.29

418. What kind of trend do the data show?

- (a) Upward (b) Downward
(c) Both upward & downward (d) No trend

419. Which component of the time series is most prominent in the data?

- (a) Seasonal Variation (b) General Trend (c) Irregular Variation (d) Cyclic Variation

Answer the next THREE questions based on the following information

Year	2016	2017	2018	2019	2020	2021	2022	2023
USD Exchange Rate	78.35	79.49	82.87	83.26	84.60	84.37	85.80	106.70

Table 4: Source–Investing.com

420. What is the second value of semi-average method?

- (a) 85.40 (b) 90.37 (c) 91.73 (d) 89.78

421. What kind of a trend do the data have?

- (a) Upward (b) Downward
(c) Both upward & downward (d) No trend

422. Which component of time series is visible in the later part of the data?

- (a) Seasonal Variation (b) General Trend (c) Irregular Variation (d) Cyclic Variation

Answer the next THREE questions based on the following information

423. What is the semi-average for the second period of the data?

- (a) 160 (b) 165 (c) 180 (d) 190

Month	January	February	March	April	May	June	July	August
Rainfall (mm)	150	120	180	200	160	140	170	190

Table 5: Source: Meteorological Department

424. Which type of trend do these rainfall data indicate?

- (a) Increasing (b) Decreasing (c) No trend (d) Fluctuating

425. What is the primary variation component observed in the data?

- (a) Seasonal Variation (b) Trend Variation (c) Cyclic Variation (d) Irregular Variation

426. Time Series has how many components?

- (a) 2 (b) 3 (c) 4 (d) 5

427. Which component involves period more than one (01) year?

- (a) Seasonal Variation (b) Cyclic Variation (c) Irregular Variation (d) Random Variation

428. Which one is not a component of Time Series

- (a) Seasonal Variation (b) Cyclic Variation (c) General Trend (d) Regular Variation

429. A company is constantly getting greater revenue than previous year; this is—

- (a) Seasonal Variation (b) General Trend (c) Irregular Variation (d) Cyclic Variation

430. Which is not a method of finding general trend?

- (a) Graphical Method (b) Moving Average (c) Semi-Average (d) Moving Median

Answer the next two questions based on the following table:

Year	2007	2008	2009	2010	2011	2012
Sales	5	35	34	40	42	204

431. In Semi-Average method, what is the 2nd average?

- (a) 74 (b) 24.67 (c) 95.33 (d) 28

432. What is the last value of 3-yearly moving average?

- (a) 93.55 (b) 95.53 (c) 95.33 (d) 59.33

433. Which component of time series is affected by economic changes due to war?

- (a) Trend (b) Seasonal Variation (c) Irregular Variation (d) Cyclic Variation

434. Which component of a time series captures long-term upward or downward movement?

- (a) Trend (b) Seasonal Variation (c) Irregular Variation (d) Cyclic Variation

435. Which time series component represents fluctuations occurring at regular intervals within a year?

- (a) Trend (b) Seasonal Variation (c) Irregular Variation (d) Cyclic Variation

436. Which component of time series is affected by economic changes during a recession?

- (a) Trend (b) Seasonal Variation (c) Irregular Variation (d) Cyclic Variation

437. Which component of time series is most likely to be impacted by weather conditions like a monsoon season?

- (a) Trend (b) Seasonal Variation (c) Irregular Variation (d) Cyclic Variation

438. Which component of time series would be influenced by government policy changes such as tax reforms?

- (a) Trend (b) Seasonal Variation (c) Irregular Variation (d) Cyclic Variation

Answer the next three questions based on the following table:

Year	2016	2017	2018	2019	2020
Car Sales	1200	1500	1700	1600	1800

439. What is the first value of the 2-yearly moving average?

- (a) 1350 (b) 1300 (c) 1400 (d) 1250

440. What is the last value of the 3-yearly moving average?

- (a) 1600 (b) 1670 (c) 1630 (d) 1750

441. What is the semi-average for the first period of the data?

- (a) 1350 (b) 1400 (c) 1450 (d) 1300

442. Demand for warm clothes is higher in winter season and less in summer. Which component of time series deals with this change?

- (a) Trend (b) Seasonal Variation (c) Irregular Variation (d) Cyclic Variation

443. Death rates of a country for 7 years are given below:

Year	2009	2010	2011	2012	2013	2014	2015
Rate	5	7	6	8	7	12	13

In semi-average method, which year will be excluded?

- (a) 2012 (b) 2013 (c) 2015 (d) 2009

444. Which component of time series represents a natural disaster?

- (a) Seasonal Variation (b) General Trend (c) Irregular Variation (d) Cyclic Variation

445. How many models of time series are there to combine the components?

- (a) 2 (b) 3 (c) 4 (d) 5

446. Which one reflects an irregular variation?

- (a) Fluctuation in production due to war (b) Price hike due to famine
(c) Rise of Temperature to drought (d) Any of the above

7.1 Situation Set

Answer the next three questions based on the following table:

447. Death rates of a country for 7 years are given below:

Year	2009	2010	2011	2012	2013	2014	2015
Rate	5	7	6	8	7	12	13

In semi-average method, what is the first average?

- (a) 5 (b) 7 (c) 6 (d) 8

448. What is the first value of the 2-yearly moving average?

- (a) 5 (b) 6 (c) 7 (d) 8

449. What is the last value of the 3-yearly moving average?

- (a) 11.10 (b) 9.68 (c) 10.65 (d) 10.67

Answer the next three questions based on the following table:

The following table shows the population growth rate (in percentage) of a city over seven years.

Year	2015	2016	2017	2018	2019	2020	2021
Rate (%)	2.5	2.7	3.1	3.6	3.9	4.2	4.5

450. What is the average population growth rate over the 7 years?

- (a) 3.2% (b) 3.5% (c) 3.6% (d) 3.8%

451. What is the second value in the 3-yearly moving average?

- (a) 2.8% (b) 3.1% (c) 3.3% (d) 3.5%

452. Using the semi-average method, what is the second average?

- (a) 3.6% (b) 3.7% (c) 3.8% (d) 4.0%

Answer the next three questions based on the following table:

The following table shows the annual rainfall (in cm) recorded in a region over seven years.

Year	2010	2011	2012	2013	2014	2015	2016
Rainfall (cm)	85	90	88	92	95	100	105

453. What is the median annual rainfall for the given years?

- (a) 90 cm (b) 92 cm (c) 93 cm (d) 95 cm

454. What is the first value of the 2-yearly moving average?

- (a) 86.5 cm (b) 87 cm (c) 88.5 cm (d) 89 cm

455. Using the semi-average method, what is the first average?

- (a) 88 cm (b) 89 cm (c) 90 cm (d) 91 cm

Answer the next three questions based on the following table:

The following table shows the average monthly temperature (in °C) recorded in a city over seven months.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul
Temperature (°C)	12	14	18	22	26	30	32

456. What is the mean temperature over the given months?

- (a) 19.5°C (b) 20.5°C (c) 21.5°C (d) 22.5°C

457. What is the third value in the 3-monthly moving average?

- (a) 16°C (b) 18°C (c) 20°C (d) 22°C

458. Using the semi-average method, what is the second average temperature?

- (a) 24°C (b) 25°C (c) 26°C (d) 27°C

Answer the next three questions based on the following table:

The following table shows the monthly sales revenue (in thousand dollars) of a store over seven months.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul
Revenue (000\$)	50	55	60	70	75	80	85

459. Which month had the highest sales revenue?

- (a) May (b) Jun (c) Jul (d) Apr

460. What is the first value of the 2-monthly moving average?

- (a) 52.5 (b) 55 (c) 57.5 (d) 60

461. Using the semi-average method, what is the first average revenue?

- (a) 57.5 (b) 55 (c) 62.5 (d) 65

7.2 Multiple Completion

462. Which of the following are components of a time series?

- i. Trend
- ii. Seasonal Variation
- iii. Correlation

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

463. Which statements about time series models are correct?

- i. The additive model adds all the components
- ii. The multiplicative model also contains some additions.
- iii. Additive and multiplicative models produce identical forecasts.

Which one is correct?

- (a) ii (b) iii (c) i (d) i, ii and iii

464. Which of the following are methods of estimating trend in time series?

- i. Moving Average Method
- ii. Sem-average method
- iii. Simple Random Sampling

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

8 Published Statistics in Bangladesh

465. Limitations of published statistics in Bangladesh are –

- i. Wrong data collection method
- ii. Insufficient data
- iii. Lack of proper training

Which one is correct?

- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

466. How many sources of published statistics are there in Bangladesh?

- (a) 2 (b) 3 (c) 4 (d) 6

467. Bangladesh Bureau of Statistics collect –

- (a) Official statistics (b) Non-official statistics(c) Semi-official statistics(d) None of the above

468. Which statistics are published by an NGO?

- (a) Official statistics (b) Non-official statistics(c) Semi-official statistics(d) None of the above

469. The primary source of official statistics in Bangladesh is –

- (a) WHO (b) BBS (c) CPD (d) UNDP

470. Which statistics are typically published by NGOs like World Wildlife Fund (WWF)?

- (a) Official statistics (b) Non-official statistics(c) Semi-official statistics(d) None of the above

471. Which organization typically publishes non-official statistics in the field of health?

- (a) UNICEF (b) World Health Organization (WHO)
(c) World Bank (d) United Nations (UN)

472. In Bangladesh, a census is usually done every – years

- (a) 20 (b) 15 (c) 10 (d) 12

473. Population census is –

- (a) Official statistics (b) Non-official statistics(c) Semi-official statistics(d) None of the above

474. In Bangladesh, which ministry present the budget?

- (a) Planning (b) Education (c) Finance (d) Agriculture

Answer Key:

1. (d) R.A. Fisher
2. (d) Database creation
3. (d) Red blood cells in a person's body
4. (c) Stars in the Milky Way
5. (b) Fish in the Pacific Ocean
6. (a) i and ii
7. (b) $\sum_{i=1}^{20} cx_i = nc \sum_{i=1}^{20} x_i$
8. (d) Regression
9. (c) Correlation
10. (c) Regression analysis
11. (b) Water molecules in the ocean
12. (a) Books in a school library
13. (b) Grains of sand on a beach
14. (d) Ordinal
15. (b) Ordinal
16. (c) Interval
17. (a) Nominal
18. (a) $y_i = \frac{x_i}{a}$
19. (c) 150
20. (a) 100
21. (c) 80
22. (a) 50
23. (c) Sample
24. (b) $b \sum_{i=1}^n x_i$
25. (c) 4
26. (d) Success rate
27. (c) Ratio scale
28. (d) Ratio
29. (d) Grade in a subject
30. (b) Number of cars in a parking lot
31. (b) Number of students in a class
32. (b) Number of books on a shelf
33. (a) Shoes sizes available in a store
34. (a) Grades on a multiple-choice test (A, B, C, D)
35. (a) Outcomes of rolling a die
36. (a) Counts of people in a room
37. (a) Number of languages spoken by a person
38. (d) No. of particles in atoms
39. (c) 206
40. (d) 122
41. (b) 65
42. (c) 42
43. (c) 54
44. (d) 45
45. (d) 84
46. (c) 8
47. (b) 62
48. (b) 6
49. (c) 90
50. (d) 435
51. (c) 2612
52. (d) 7264
53. (c) 344
54. (b) 330
55. (c) 24
56. (d) 50
57. (a) 108
58. (b) 174
59. (a) i and ii
60. (a) Temperature
61. (c) Gender
62. (c) Educational Level
63. (a) Temperature
64. (c) Ratio scale
65. (d) Grade in a subject
66. (a) $\prod x_i^2$
67. (b) Continuous variable
68. (c) Mean monthly income in a city is 60,000 taka
69. (d) 13
70. (c) 93
71. (c) 99
72. (d) 119
73. (d) -34
74. (a) Room no.
75. (d) No. of member in a family
76. (c) Nominal
77. (b) 155
78. (a) 225
79. (c) 37
80. (b) 33
81. (a) 20
82. (b) 504
83. (c) 82
84. (a) 71
85. (d) 24
86. (c) 66
87. (a) 74
88. (b) 74
89. (c) 476
90. (a) 61
91. (c) 45
92. (d) 115
93. (d) 2
94. (a) Data
95. (a) Primary data
96. (c) $\theta_i = \frac{f_i}{N} \times 360$
97. (d) John Tukey
98. (b) Sample

99. (a) $K = 1 + 3.322 \log N$ 124. (a) i and ii 149. (b) Geometric Mean 173. (b) 15
100. (b) Bar Diagram 125. (a) i and ii 150. (c) Median 174. (a) 4 times
101. (c) Simple Bar Diagram 126. (a) i and iii 151. (c) Median 175. (b) $\bar{x}$
102. (a) Line Diagram 127. (a) i and iii 152. (b) Harmonic mean 176. (a) $\bar{x}$
103. (c) Ogive 128. (d) i, ii and iii 153. (d) Mode 177. (a) 40
104. (b) Component parts of a total 129. (a) i and ii 154. (b) $AM \times HM = GM^2$ 178. (c) 5.5
105. (b) Scatter Plot 130. (a) i and ii 155. (b) 6.67 179. (b) 13
106. (b) Histogram 131. (a) i and ii 156. (b) 18 180. (c) 109
107. (a) Multiple Bar Diagram 132. (a) Quartiles are well defined 157. (a) 7.07 181. (c) 109
108. (c) Ogive 133. (b) Mode 158. (c) ii and iii 182. (a) 26
109. (c) 36 134. (c) Range 158. (a) {10, 12, 14} 183. (d) 10
110. (b) 45 135. (b) When all the values are equal 159. (a) $Mo = 3Me - 2\bar{X}$ 184. (a) 20
111. (a) 44% 136. (c) Geometric Mean 160. (b) 67.5 185. (b) 20
112. (d) 6 137. (d) 5 161. (c) 44 186. (a) 88.36
113. (b) 31 138. (d) Mode 162. (b) 34 187. (a) 0
114. (a) 31 139. (b) Geometric Mean 163. (b) Positively skewed 188. (a) $\bar{X} = \frac{\sum f_i x_i}{\sum f_i}$
115. (d) 29 140. (c) 7.5 164. (b) 82.5 189. (c) 47
116. (a) 50 141. (b) 8 165. (b) 56 190. (b) n
117. (b) 45 142. (d) Mode 166. (b) Mean - Mode = 3(Mean - Median) 191. (b) $n + 1$
118. (b) 75% 143. (d) 110th Percentile 167. (c) 80 192. (b) $\frac{n+1}{2}$
119. (a) 55 144. (a) $\sum_{i=1}^n (X_i - Median)^2$ 168. (d) 36 193. (c) 32.00
120. (c) 65 145. (b) Geometric Mean 169. (c) 65 194. (a) $\frac{n}{\sum_{i=1}^n \frac{f_i}{x_i}}$
121. (c) 60% 146. (a) All values are equal 170. (c) 7 195. (c) 14.39
122. (d) 0.35 147. (b) Median 171. (c) Multiplied by 3 and then increased by 4 196. (c) Harmonic Mean
123. (d) Ogive 148. (b) Geometric Mean 172. (b) 6 197. (c) Harmonic Mean

198. (c) Reciprocal of Mean $\frac{1}{\bar{x}}$ 223. (c) i and ii 248. (c) Non-negative 273. (a) i and ii
199. (b) 1, 2, 4, 8, 16, 32 224. (c) i & ii 249. (b) $\frac{SD}{Mean} \times 100$ 274. (b) i and iii
200. (c) 5.66 225. (c) ii and iii 250. (c) 20% 275. (d) To uniquely identify a distribution
201. (b) Geometric Mean 226. (c) ii and iii 251. (b) Datasets with different units or means 276. (b) Raw moments
202. (d) 10.5 227. (a) i and ii 252. (b) Median 277. (c) (-3, 3)
203. (b) 5.66 228. (a) i and ii 253. (c) 10 278. (d) Rectified Moments
204. (a) Histogram 229. (b) i and iii 254. (d) The extreme 25% in both tails 279. (a) $\frac{\sum (x_i - \bar{x})^n}{w}$
205. (c) 8 230. (a) i and ii 255. (b) Range 280. (b) $\mu'_1 = \bar{x} - a$
206. (b) 4 231. (a) i and ii 256. (c) (6, 6) 281. (a) $\frac{\sum f_i (x_i - a)^r}{n}$
207. (c) The value that occurs most frequently 232. (a) i and ii 257. (a) 1.5 282. (c) Moments
208. (d) 5 233. (b) i and iii 258. (a) $\sqrt{\frac{1}{n} \sum (x_i - \bar{x})^2}$ 282. (d) i, ii and iii
209. (c) Mean = Median = Mode 234. (d) i, ii and iii 259. (b) $\sqrt{\frac{(n^2-1)}{12}}$ 283. (c) μ_2
210. (c) Median 235. (a) i and ii 260. (c) 6 284. (c) $\frac{\sqrt{\mu_2}}{\bar{x}} \times 100$
211. (c) Ogive 236. (a) i and ii 261. (a) 2.87 285. (d) Arithmetic Mean
212. (b) 44 237. (b) Standard deviation 262. (d) Coefficient of variation 286. (c) First central moment
213. (c) 30 238. (a) 45 263. (d) Quartile deviation 287. (b) μ_3
214. (d) 53.75 239. (b) 0 264. (b) 2.5 288. (d) $\mu'_2 - \mu_1'^2$
215. (b) 32 240. (b) $\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2$ 265. (d) 7.7% 289. (b) 0
216. (b) 25-50 241. (c) $V(X) = V(Y)$ 266. (c) ii and iii 290. (d) $\bar{x} - a$
217. (c) 3.5 242. (c) -10, 10, 30 267. (a) i and ii 291. (b) -2
218. (b) 70 243. (c) Standard deviation 268. (d) i, ii and iii 292. (c) 0
219. (d) 74 244. (c) 0 269. (d) i, ii and iii 293. (a) 2
220. (d) 70th percentile 245. (b) (4,4) 270. (d) i, ii and iii 294. (a) 2
221. (b) 70 246. (d) 16 271. (d) i, ii and iii 295. (a) 10
222. (c) 51.5 247. (b) Becomes 3 times 272. (a) i and ii 296. (b) -3.4
273. (a) i and ii 297. (a) 0 298. (a) 3.2

299. (c) 5	325. (c) 3	350. (a) $Q_1 - 1.5 \times IQR$	375. (a) -0.92
300. (b) 20	326. (c) 8.25	351. (b) $IQR = Q_3 - Q_1$	376. (d) -1
301. (d) 21	327. (c) Symmetric	352. (a) Mode	377. (a) Correlation
302. (c) 60	328. (a) Positively skewed	353. (d) i, ii & iii	378. (c) Correlation coefficient
303. (b) i and iii	329. (b) Negatively skewed	354. (b) 50	379. (a) Qualitative variable
304. (b) 11.92	330. (a) i and ii	355. (a) i and ii	380. (d) 0
305. (d) 110	331. (c) ii and iii	356. (b) i and iii	381. (c) Perfect positive correlation
306. (b) Symmetrical	332. (b) i and iii	357. (d) i, ii and iii	382. (a) Perfect negative correlation
307. (a) Positive Skew	333. (a) $\gamma_1 = \sqrt{\frac{\mu_3^2}{\mu_2^3}}$	358. (a) Arithmetic Mean	383. (b) Partial positive correlation
308. (a) 0	334. (b) 20	359. (b) 4	384. (d) Partial negative correlation
309. (b) Negative Skew	335. (d) 58	360. (b) -0.5	385. (d) Zero correlation
310. (a) Positive Skew	336. (a) 52.67	361. (b) 3.0	386. (a) Choice
311. (d) All of the above	337. (a) 15	362. (c) -1 and +1	387. (d) -0.94
312. (b) Median	338. (c) 26	363. (c) Both change of origin and scale	388. (d) -0.94
313. (c) Mode	339. (d) i, ii and iii	364. (d) Strong negative correlation	389. (c) Cannot be determined
314. (a) Mean	340. (a) A	365. (a) 0.6	390. (b) 0.60
315. (b) Positively skewed	341. (b) 3	366. (a) Two regression coefficients	391. (c) 0.67
316. (c) $M_o = 3Me - 2\bar{x}$	342. (d) 48	367. (a) 1	392. (a) i and ii
318. (b) leptokurtic	343. (c) Mesokurtic	368. (c) 0	393. (c) No. of earthquakes in different regions
319. (d) 29.45	344. (a) Leptokurtic	369. (c) Data is in ranks or ordinal	394. (c) Number of students in a each class
320. (a) $Mean > Median > Mode$	345. (c) Mesokurtic	370. (b) 0.75	395. (a) Number of calls received by a call centre
321. (c) 0	346. (b) Platykurtic	371. (a) -1 and +1	396. (a) i and ii
322. (a) Left Skew	347. (b) $\gamma_2 = \beta_2 - 3$	372. (b) 0.63	397. (c) Moving average method
323. (c) Right Skew	348. (c) 3	373. (c) Uncorrelated	398. (a) $Y_t = T_t \times S_t \times C_t \times R_t$
324. (b) Symmetry	349. (d) $\gamma_2 = \beta_2 - 3$	374. (d) Karl Pearson	399. (a) 0
			400. (b) 1
			401. (d) Seasonal variation
			402. (b) 190

403. (a) a curved line	421. (a) Upward	439. (a) 1350	457. (b) 18°C
404. (a) Increase in ice cream sales during summer	422. (c) Irregular Variation	440. (c) 1630	458. (c) 26°C
405. (b) A toy store	423. (b) 165	441. (a) 1350	459. (c) Jul
406. (a) Boom and recession phases fluctuating	424. (b) Seasonal Variation	442. (b) Seasonal Variation	460. (a) 52.5
407. (a) Gradual increase in global temperature	425. (a) Seasonal Variation	443. (b) 2013	461. (b) 55
408. (a) Upward trend	426. (c) 4	444. (c) Irregular Variation	462. (a) i and ii
409. (a) Declining birth rate in developed countries	427. (b) Cyclic Variation	445. (c) 2	463. (c) i
410. (a) Improved marketing strategy	428. (d) Regular Variation	446. (d) Any of the above	464. (a) i and ii
411. (b) Long-term business cycle	429. (a) General Trend	447. (c) 6	465. (d) i, ii and iii
412. (d) Any of the above	430. (d) Moving Median	448. (b) 6	466. (b) 3
413. (b) Graphical method	431. (c) 95.33	449. (c) 10.65	467. (a) Official statistics
414. (b) 90.37	432. (c) 95.33	450. (b) 3.5%	468. (c) Semi-official statistics
415. (a) Upward	433. (c) Irregular Variation	451. (b) 3.1%	469. (b) BBS
416. (c) Irregular Variation	434. (a) Trend	452. (c) 3.8%	470. (b) Non-official statistics
417. (c) 25.88	435. (b) Seasonal Variation	453. (b) 92 cm	471. (b) World Health Organization (WHO)
418. (a) Upward	436. (c) Irregular Variation	454. (a) 86.5 cm	472. (c) 10
419. (b) General Trend	437. (b) Seasonal Variation	455. (b) 89 cm	473. (a) Official statistics
420. (b) 90.37	438. (d) Cyclic Variation	456. (c) 21.5°C	474. (c) Finance